



Anik Shekhar Mazumder

CIVIL & ENVIRONMENTAL
ENGINEER

CONTACT

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- Sylhet, Bangladesh
- Anik Shekhar Mazumder
- My-Portfolio

TECHNICAL SKILLS

Structural Design	85%
GIS & Spatial Analysis	90%
Environmental Impact	85%
Sustainability/LCA	80%
Python & Automation	75%
Project Management	78%

LANGUAGES

Bangla	Native
English	Fluent
German	Basic

Anik Shekhar Mazumder

B.SC. CIVIL & ENVIRONMENTAL ENGINEERING · SUST, SYLHET ·
FINAL SEMESTER

PROFESSIONAL SUMMARY

Detail-oriented Civil & Environmental Engineering graduate (final semester, CGPA 3.01) at Shahjalal University of Science and Technology, with demonstrated expertise in structural design, GIS-based spatial analysis, and environmental life cycle assessment. Adept at applying Python-driven automation to real-world engineering challenges with a strong research mindset focused on sustainable infrastructure and urban waste management. Seeking opportunities in graduate research, international internships, or entry-level engineering roles where technical rigor and environmental stewardship intersect.

EDUCATION

B.Sc. Engineering — Civil & Environmental Engineering

2020 – Present

Shahjalal University of Science and Technology (SUST)
Sylhet, Bangladesh · CGPA: 3.01 · Final Semester (4/2) — Ongoing

Higher Secondary Certificate (HSC) — Science

2018 – 2020

Cumilla Victoria Government College
Cumilla, Bangladesh · GPA: 5.00 / 5.00

Secondary School Certificate (SSC) — Science

2016 – 2018

Ispahani Public School and College
Cumilla, Bangladesh · GPA: 5.00 / 5.00

SOFTWARE & TOOLS

AutoCAD

ETABS

ArcGIS Pro

OpenLCA

MODFLOW

Python

C Programming

MS Office Suite

Web Scraping

Front-End Development

RESEARCH & PROJECTS

Undergraduate Thesis — GIS-Based Life Cycle Assessment of Plastic Waste-to-Energy Conversion via Pyrolysis

Nov 2025 – Present

A Sustainable Framework for Urban Waste Management in Sylhet City

- ▶ Conducted ward-wise spatial analysis of plastic waste generation in Sylhet using ArcGIS Pro, integrating population density and municipal waste datasets to identify high-priority waste zones.
- ▶ Performed multi-criteria site suitability analysis to determine optimal locations for waste-to-energy pyrolysis plants, applying spatial constraint modelling and overlay techniques.
- ▶ Automated geospatial data processing workflows using custom Python scripts integrated within ArcGIS Pro, significantly reducing manual processing time and improving analytical accuracy.
- ▶ Executing a comparative Life Cycle Assessment (LCA) using OpenLCA to evaluate the environmental footprint of three distinct pyrolysis conversion routes, aiming to identify the most sustainable processing pathway.
- ▶ Developing a scalable, data-driven sustainability framework for urban plastic waste management applicable to rapidly urbanising cities across South Asia.

RESEARCH INTERESTS

Sustainable Infrastructure

GIS & Geospatial Engineering

Life Cycle Assessment (LCA)

Waste-to-Energy Systems

Environmental Impact Analysis

Groundwater Modelling

Urban Resilience & Planning

Circular Economy

CERTIFICATIONS & TRAINING

- AutoCAD for Civil Engineering — Technical Drafting & 2D/3D Modelling
- ArcGIS Pro Spatial Analysis — GIS-Based Environmental Planning
- Python for Engineers — Data Processing & Automation (Self-Directed)
- ETABS Structural Modelling — Reinforced Concrete & Frame Analysis

ACHIEVEMENTS

- Achieved perfect GPA of **5.00/5.00** in both SSC and HSC national examinations — demonstrating consistent academic excellence prior to university.
- Independently designed and executed a novel GIS-LCA integrated research framework as an undergraduate thesis, combining geospatial data science with environmental engineering.
- Developed proficiency in Python scripting and front-end development alongside a core civil engineering curriculum, demonstrating multidisciplinary initiative.